

Summary

Proxima Fusion presents a **novel quench protection concept [1]** that

- **Circumvents high voltages** during a quench/fast discharge scenario
- Promises a more **uniform energy deposition** due to **>10m/s normal zone propagation velocities** (NZPV) on stack level

Simulation suggests normal zone propagation velocities of >10m/s is achievable with a stack of 5 ReBCO tapes. An experiment is being built to study this effect.

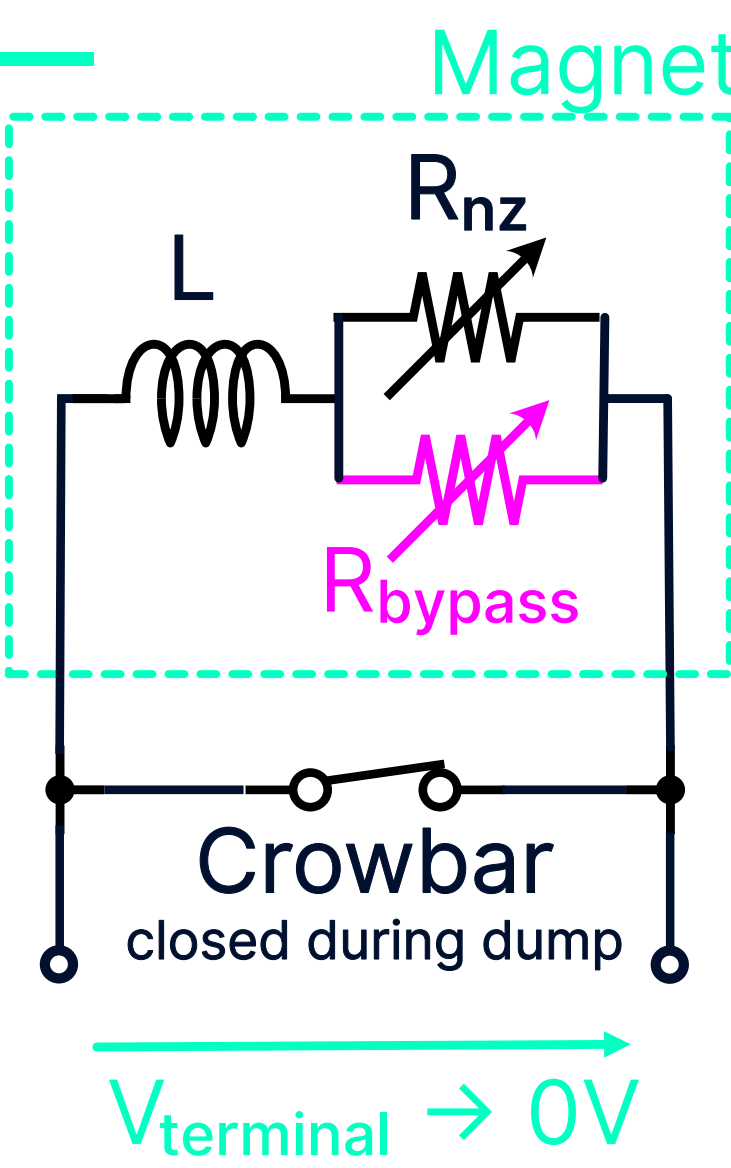
1 No terminal voltage during quench

$$L \frac{dI}{dt} = -R_{nz}(t)I, \quad V_{\text{term}} = 0$$

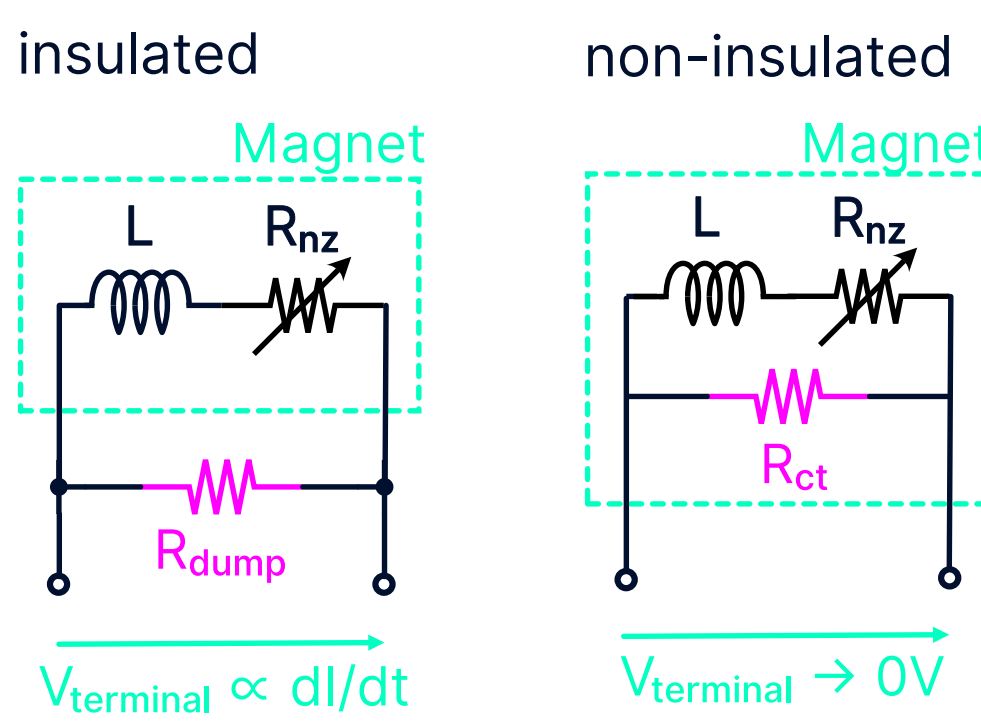
No terminal voltage - inductive voltage equals voltage drop over conductor, independent of the stored energy.

$$Al_{nz} \int_{T_{op}}^{T_{\text{safe}}} C(T) dT \geq \frac{1}{2} LI^2$$

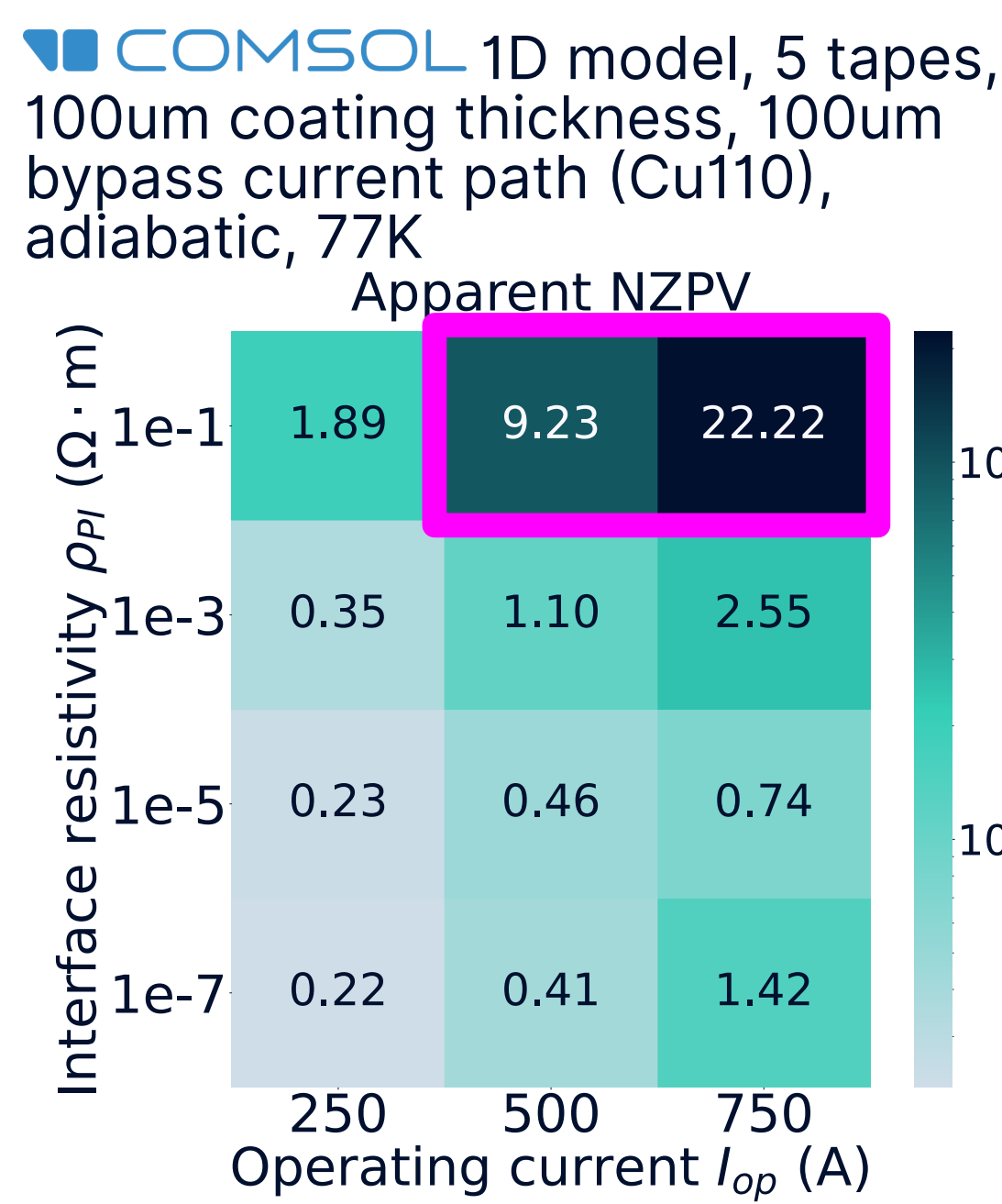
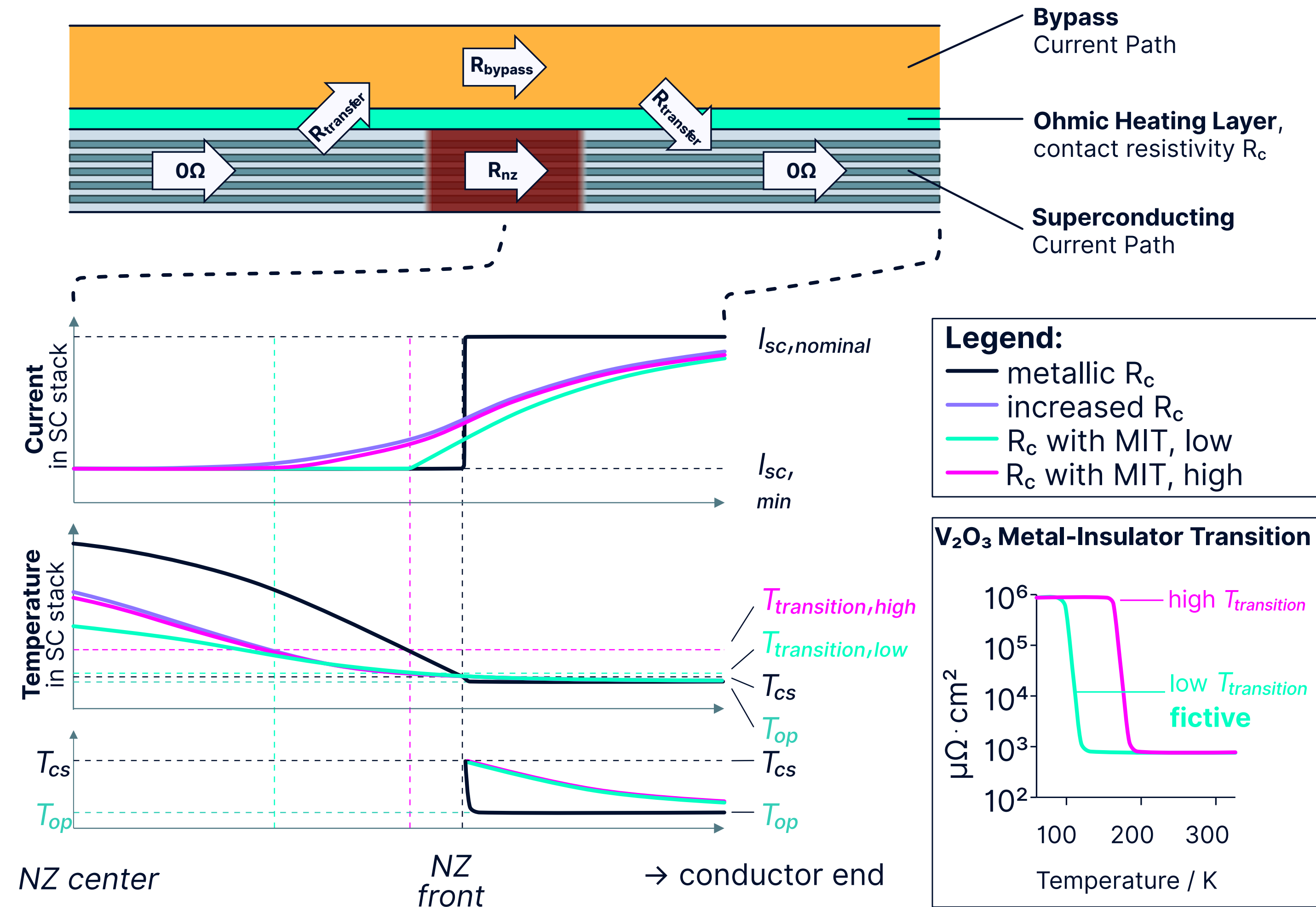
Total **magnetic energy is transferred to conductor enthalpy**. Only possible if the turn quenches uniformly. **This requires fast NZPV.**



In Comparison:



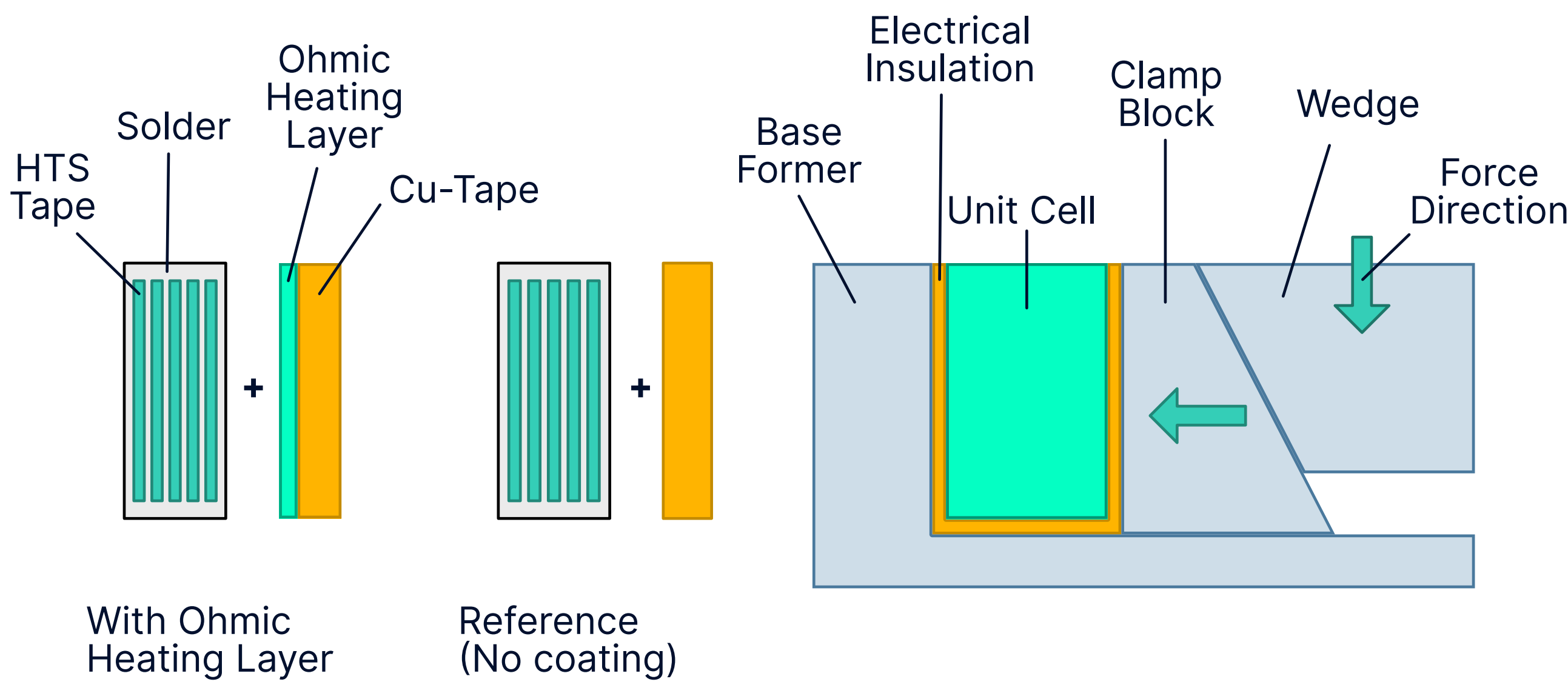
2 Normal zone propagation velocity > 10 m/s



3 Test with 1d physics instead of a coil

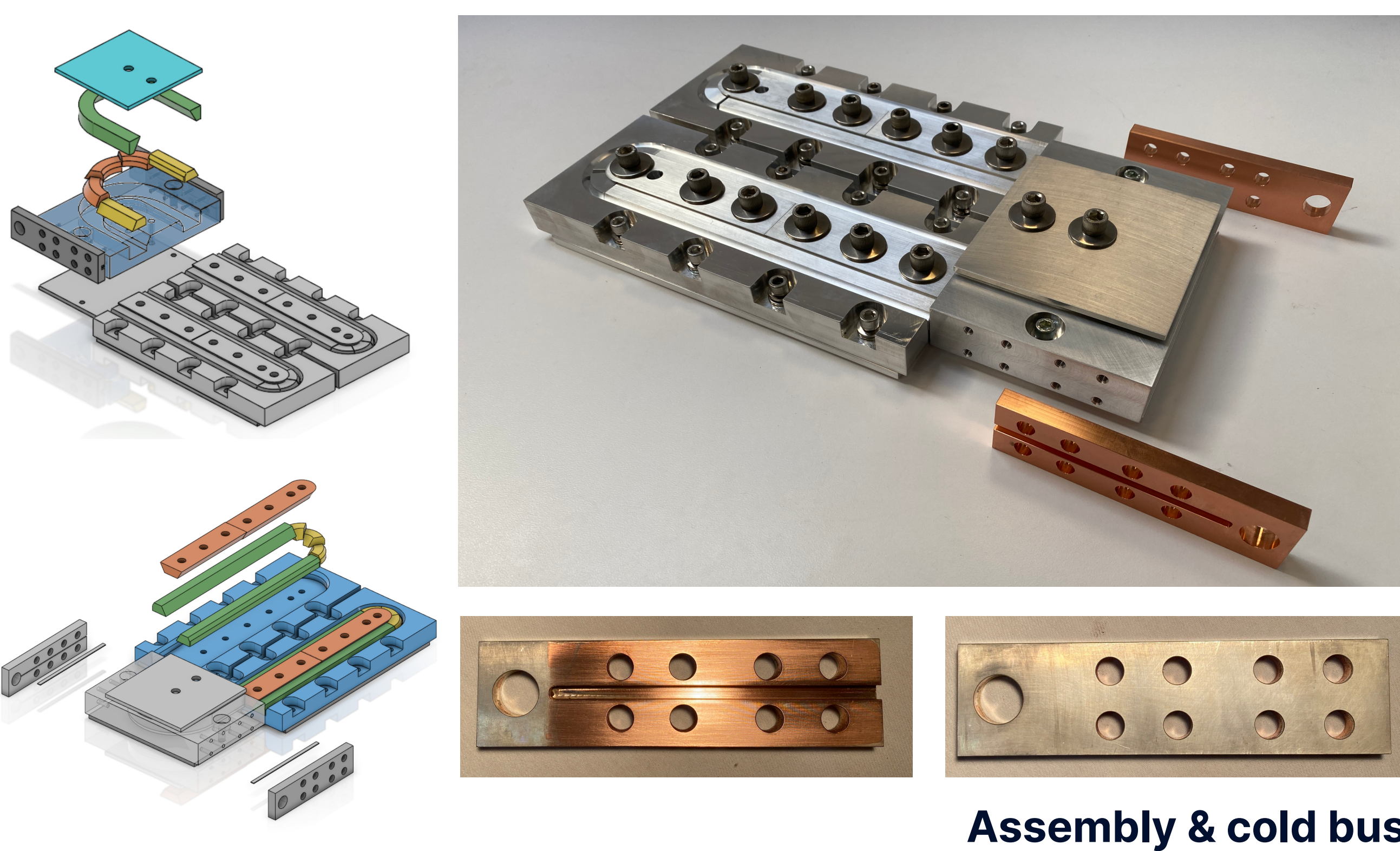
Experiment setup

- Contact pressure from same direction relative to tape
- Quasi-adiabatic
- T_{peak} < T_{safe}=400K



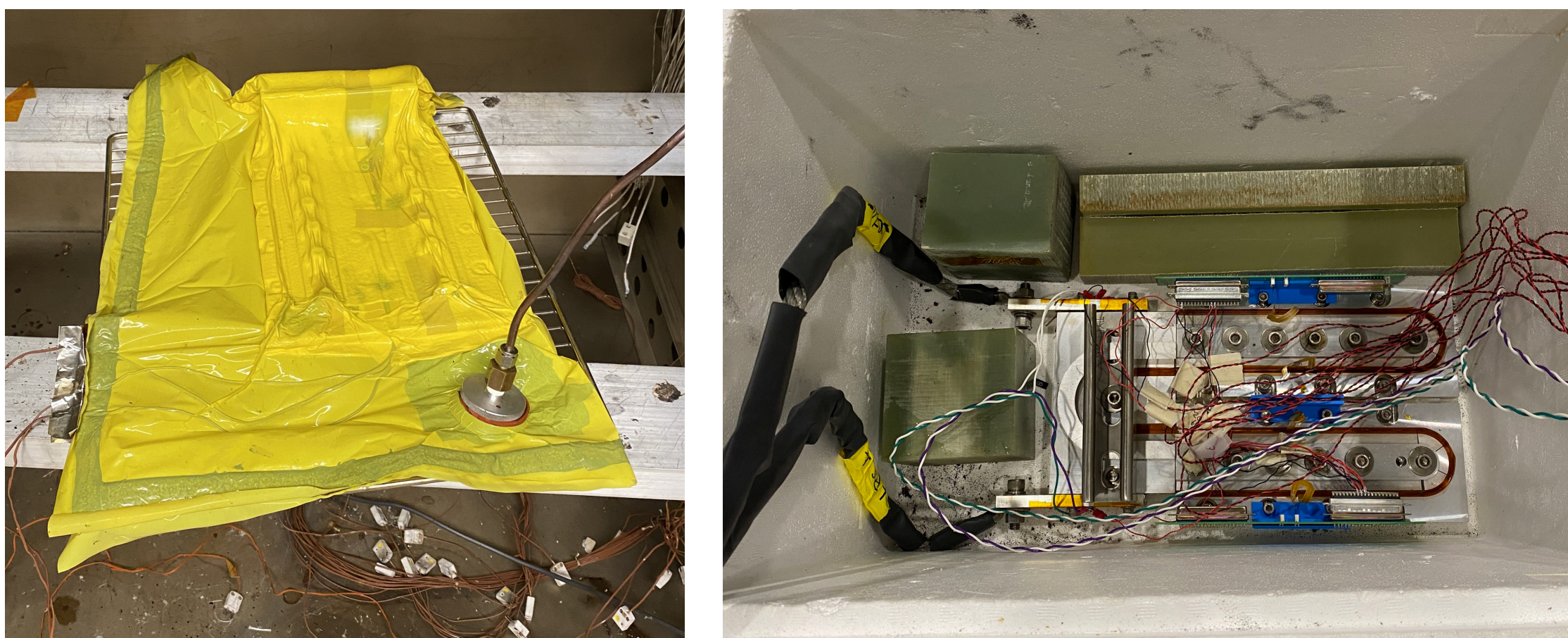
Section view of fast NZPV conductor & reference

- Materials: Alu 6061 & Cu110
- Belleville washers for sustained pressure during temperature cycles
- Torque deduced from FUJI film pressure distribution

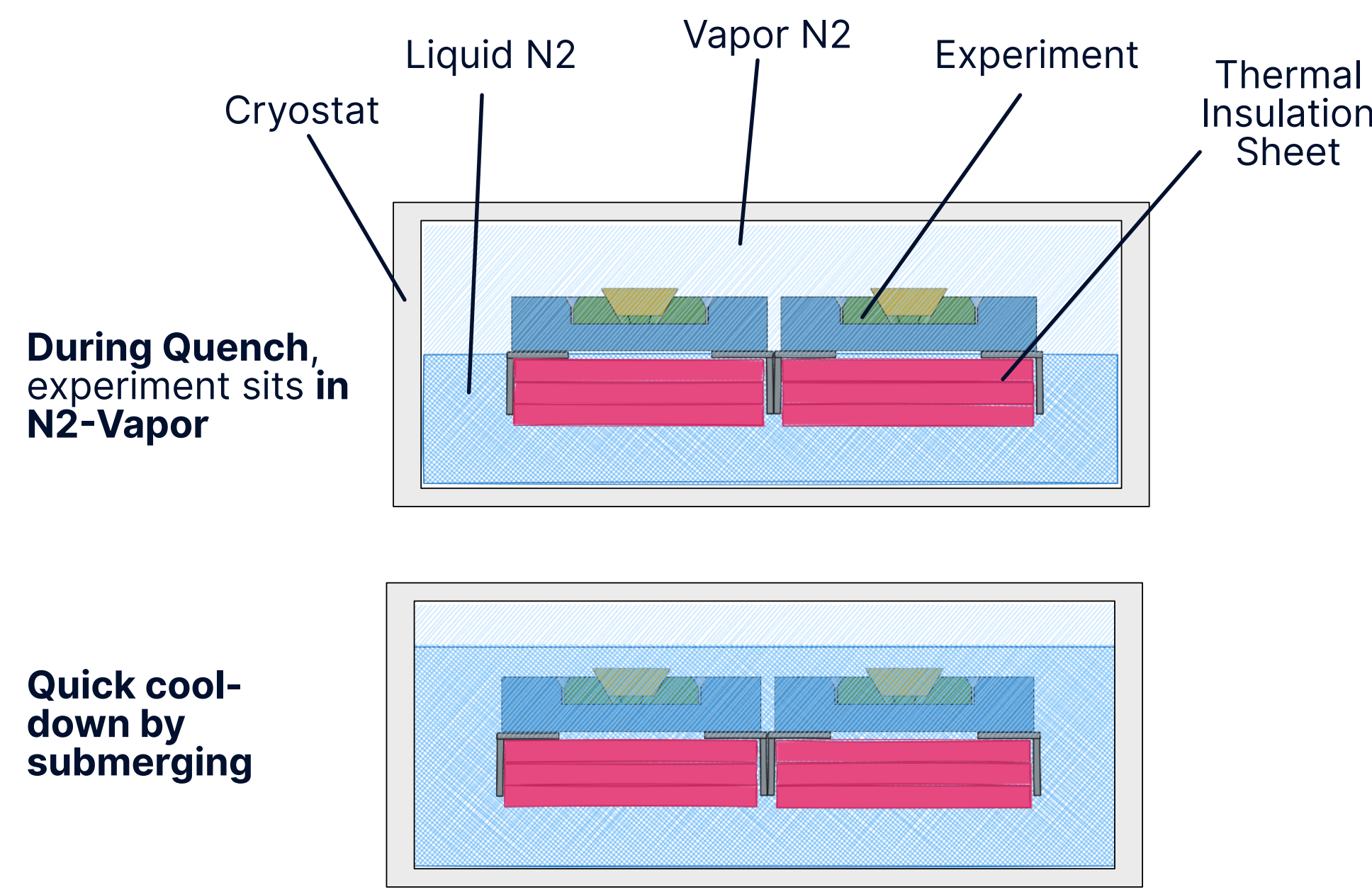


Cold bus terminals

- 40 HTS tapes each, precise current injection into conductor
- Silver-plated for lowest current transfer length

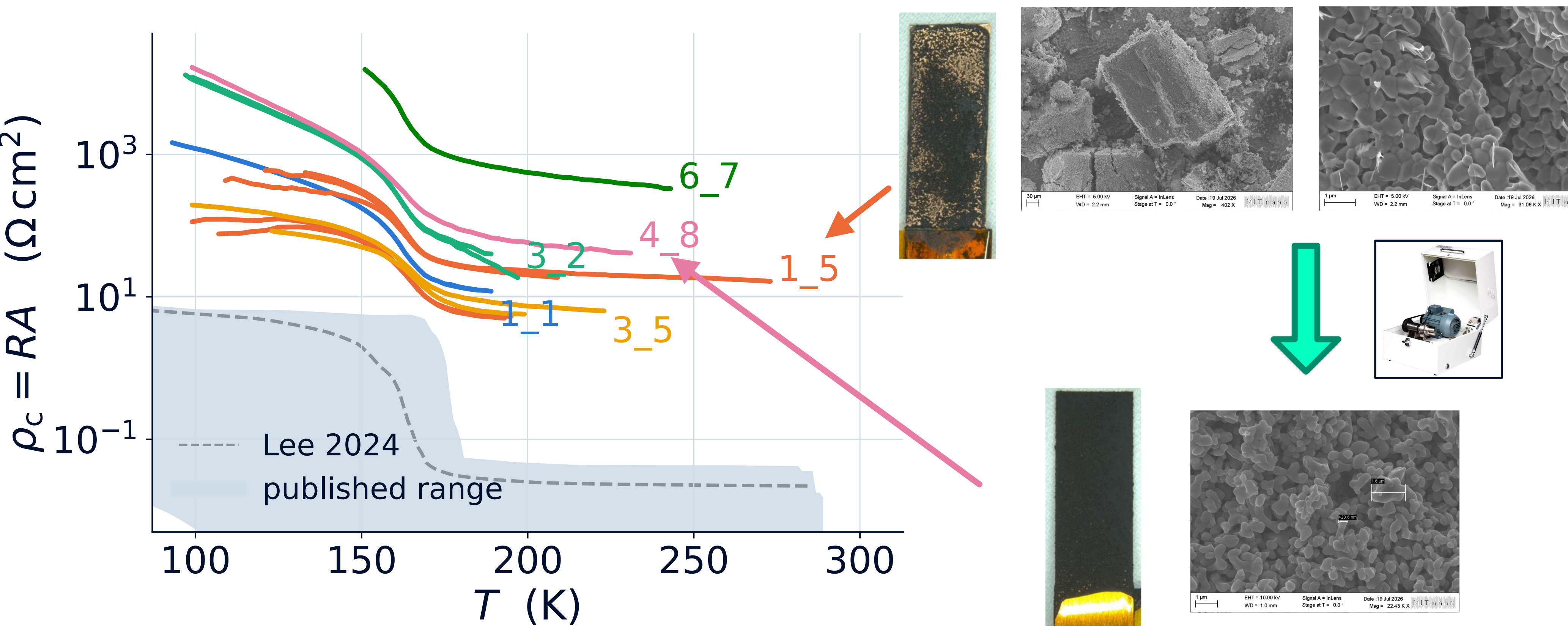


Re-flow solder process & instrumented assembly same former can be used for soldering and testing



Experiment operations fast cool-down by dunking into LN2 to allow for low downtime (<2 min) between quenches

4 Too high coating resistivity increases experiment complexity

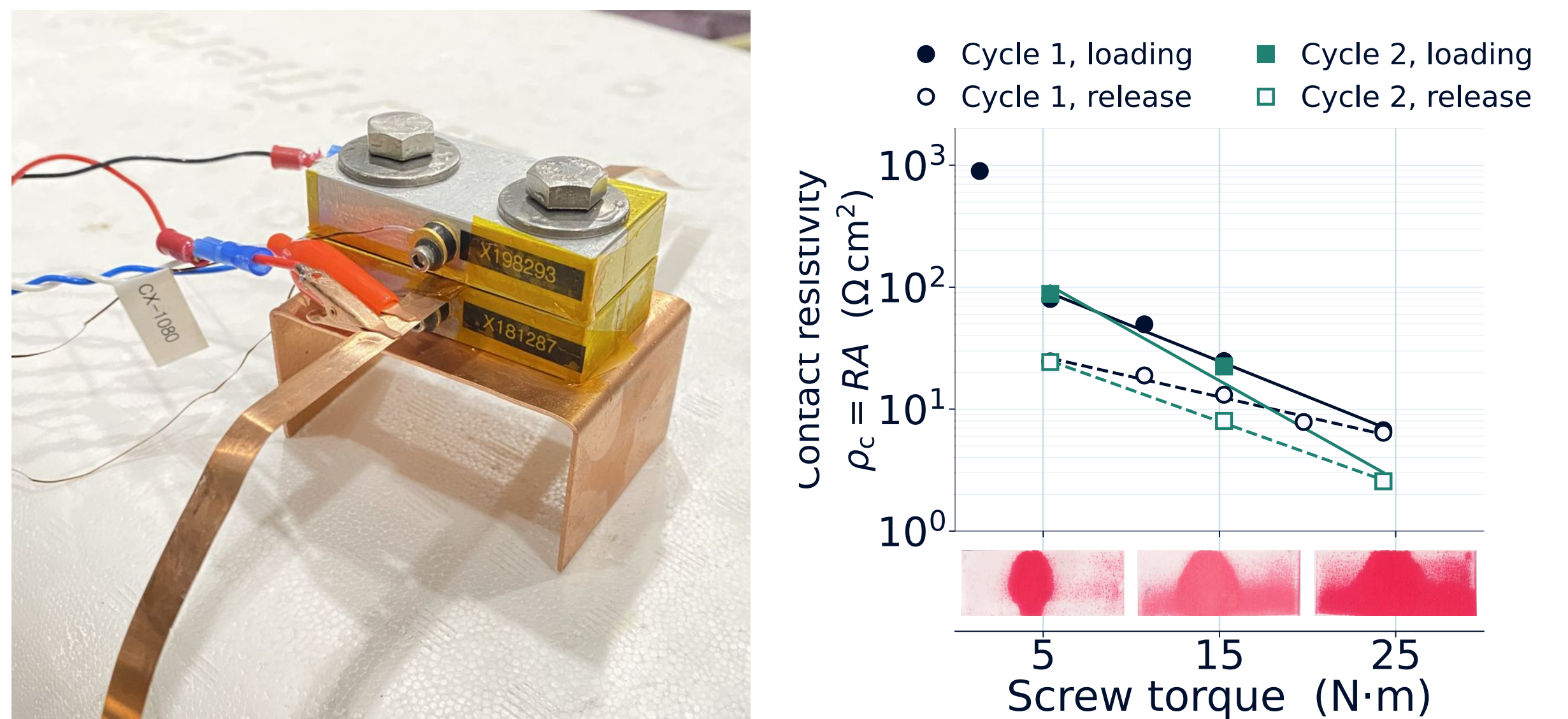


Coupon tests of coating for prototyping

Lower coating resistance decreases minimum conductor length. References [4-6]

Tuning resistivity is a challenge

- Currently achievable coating resistivity too high. Sample length of >40m.
- Coating thickness cannot regulate resistivity enough, but improves hiding power
- Doping V₂O₃ powder with less resistive material not yet successful
- Pressure regulates resistivity



Contact Pressure regulates resistance reliably Coating resistance can be adjusted by 1 OOM

Acknowledgement

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References

- [1] Proxima Fusion GmbH, P175858WO, 17 Nov 2025.
- [2] G. A. Levin, K. A. Novak and P. N. Barnes, Supercond. Sci. Technol. 23, 014021 (2010).
- [3] C. Lacroix and F. Sirois, Supercond. Sci. Technol. 27, 035003 (2014).
- [4] S. H. Lee et al., IEEE Trans. Appl. Supercond. 34, 1–9 (2024).
- [5] M. M. Mussa et al., Curr. Appl. Phys. 56, 24–35 (2023).
- [6] A. Vaskuri et al., IEEE Trans. Appl. Supercond. 33, 1–4 (2023).

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